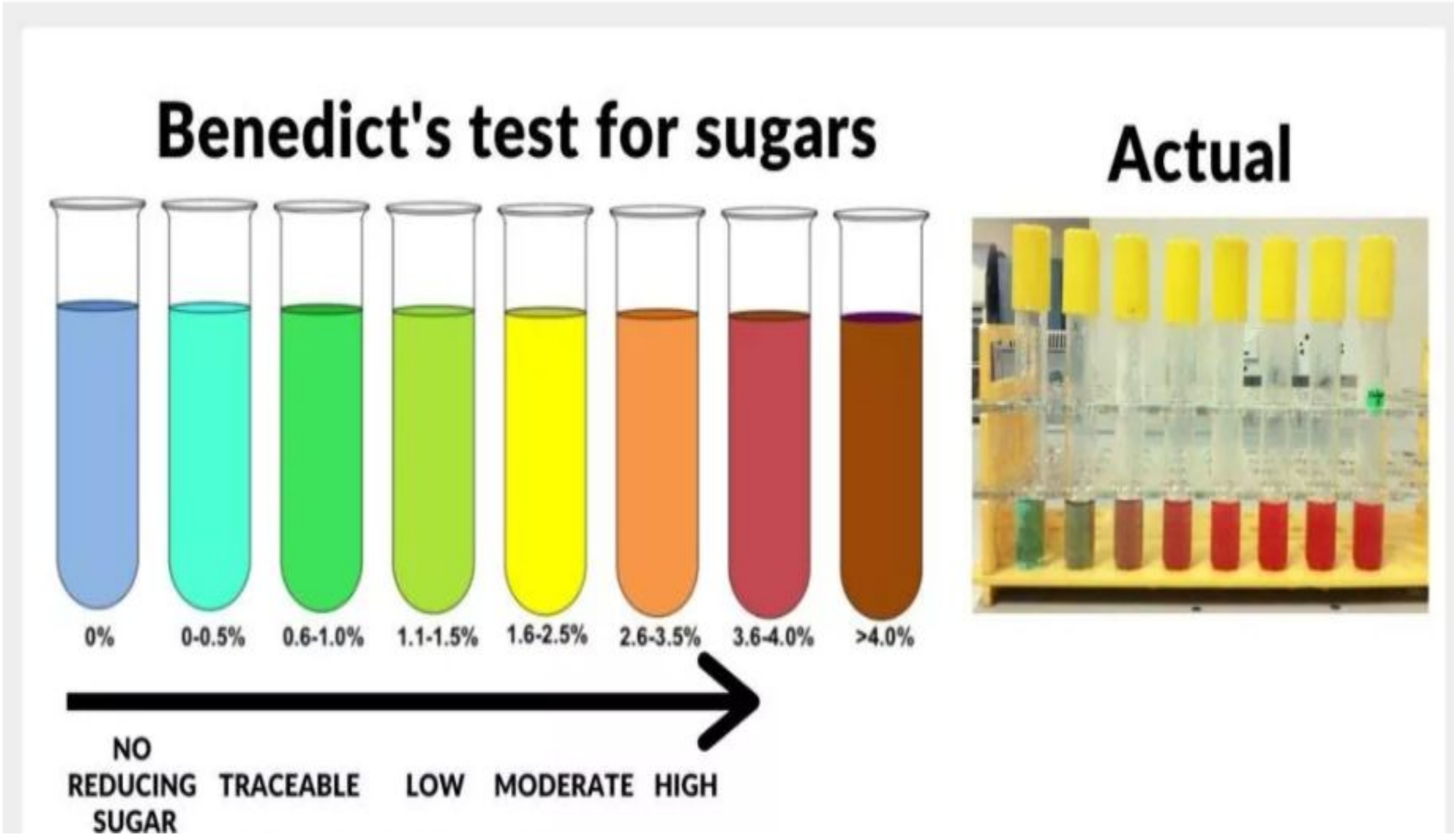
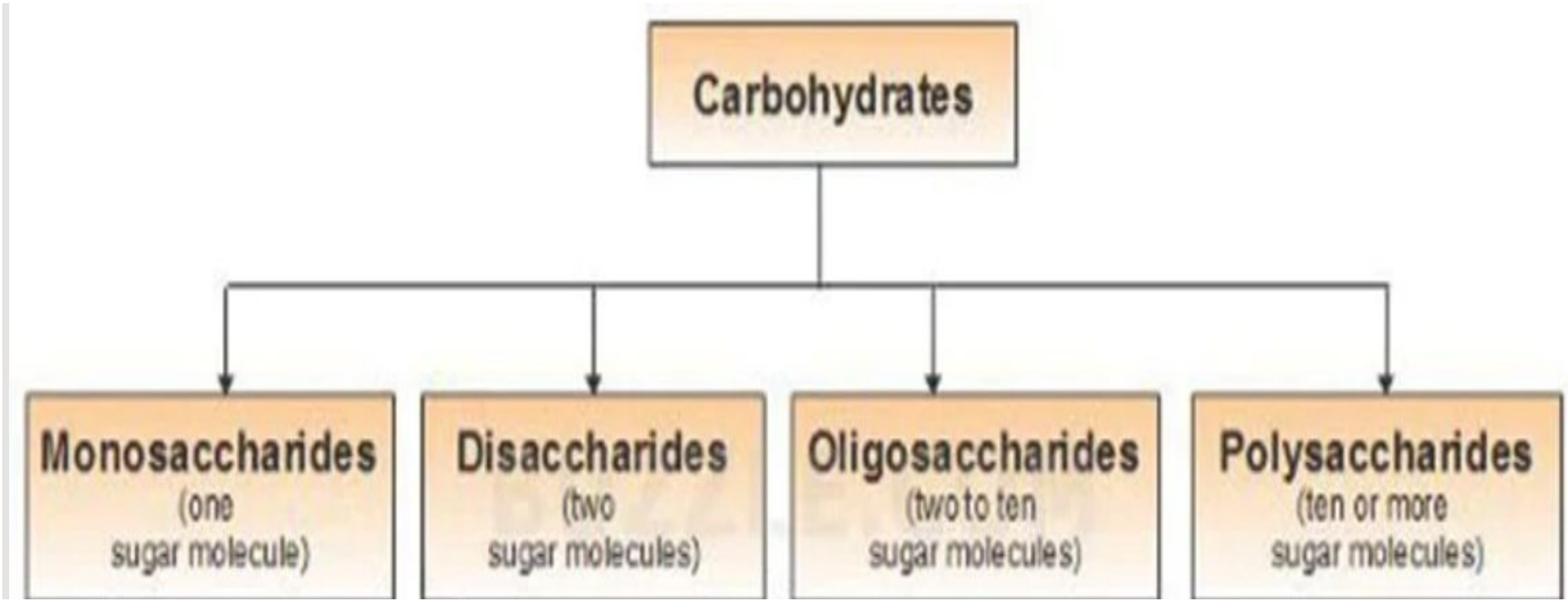


Dr. Humaira Aman Ali

CARBOHYDRATES

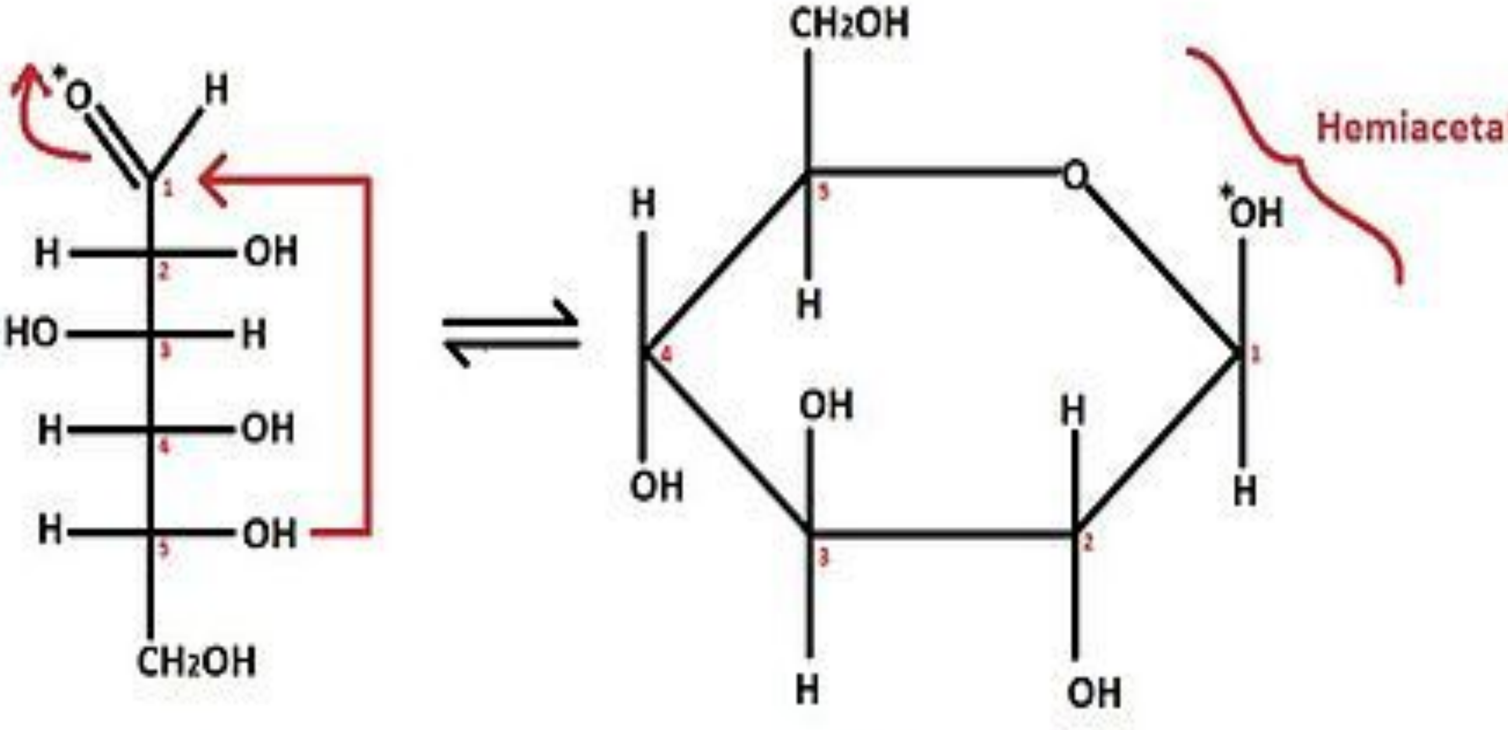
- Definition:
  - Carbohydrates can be defined as: “Polyhydroxylated compounds with at least three carbon atoms, with potentially active carbonyl groups”.

Classification of Carbohydrates



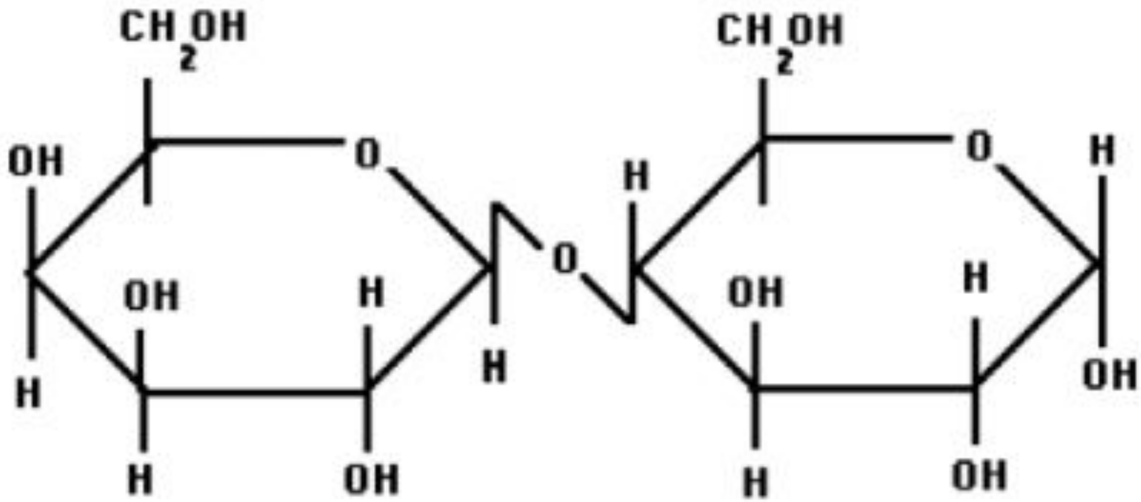
REDUCING SUGAR

- Some sugars such as glucose are called reducing sugars because they are capable of transferring hydrogens (electrons) to other compounds, a process called reduction.



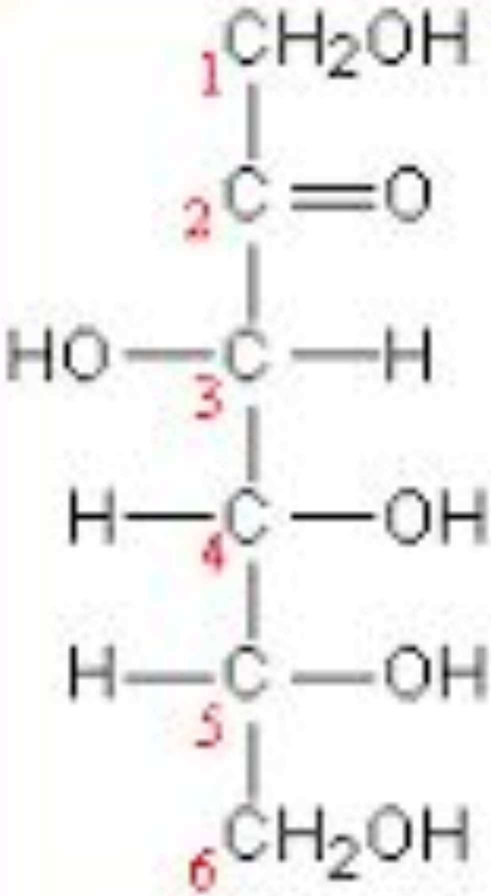
Fischer Projection of Glucose

Glucopyranose

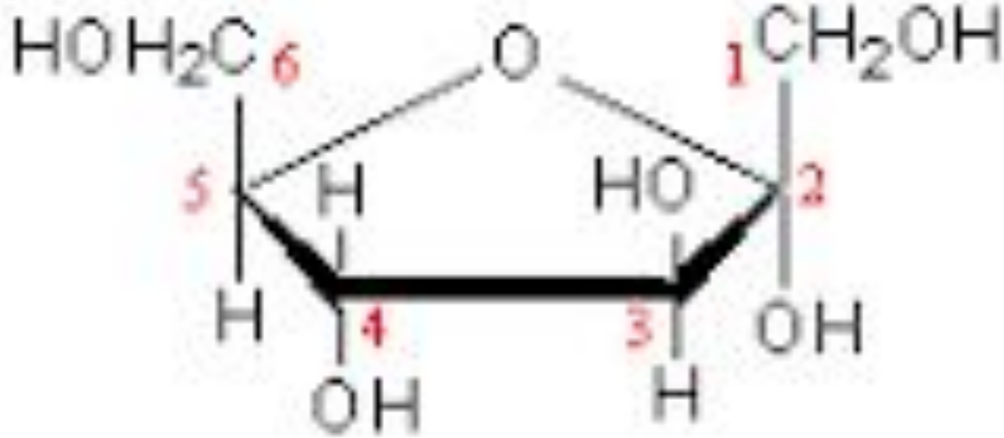


α

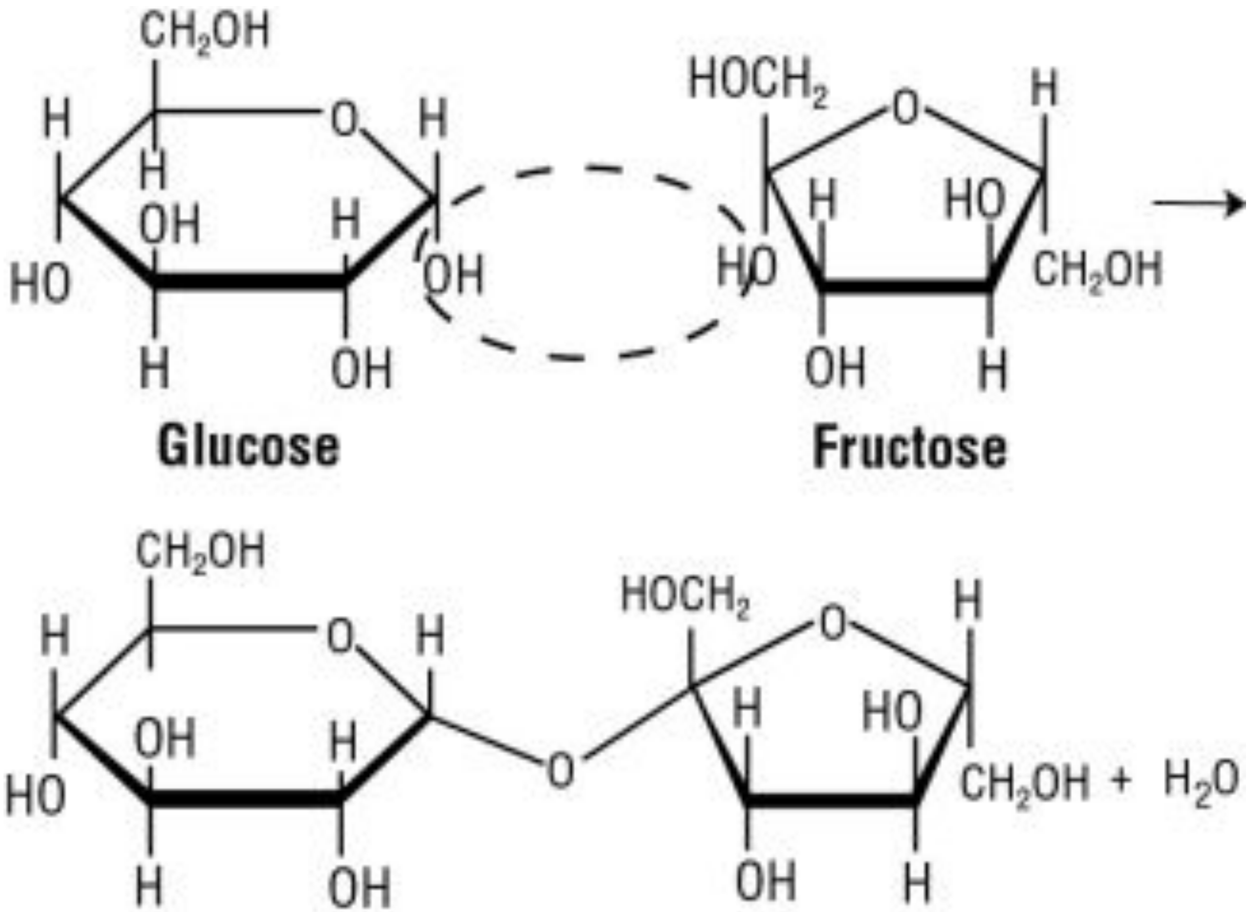
Galactose ——— Glucose



D-fructose (linear)

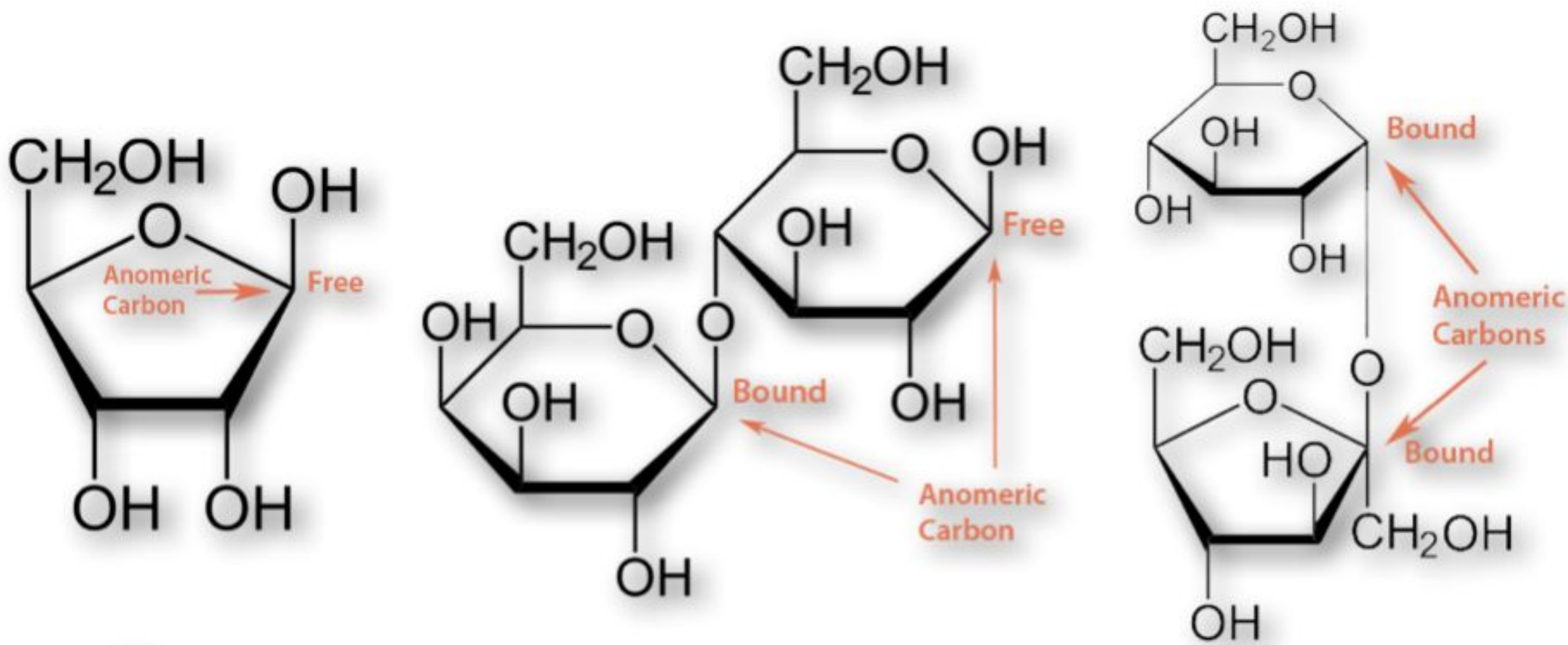


α-D-fructofuranose



Sucrose

Reducing Versus Non-Reducing Sugars



Ribose
(Reducing)

Lactose
(Reducing

Sucrose
(Non-Reducing)

SCHEME

Identification of Carbohydrates present in a given sample

MOLISCH'S TEST

POSITIVE
Carbohydrate present

NEGATIVE
carbohydrate absent

IODINE TEST

POSITIVE
Polysaccharide
Blue - starch
Purple - Dextrin
Reddish- glycogen

NEGATIVE
Monosaccharide / Disaccharide

BENEDICTS TEST

POSITIVE
Reducing sugar

NEGATIVE
non reducing disaccharides

Monosaccharide –e.g glucose
Fructose

Disaccharide – e.g. Maltose
Lactose

e.g. sucrose.
C.T
Thymol blue test

BARFOED TEST

BARFOED TEST

POSITIVE
With in3minutes
Monosaccharide present

NEGATIVE
more than 3minutes
reducing disaccharide present

SELIWANOFF'S TEST

POSITIVE

With in 30 sec
ketosugar(fructose)

NEGATIVE

after 30 sec
aldo sugar(glucose)

C.T
OSAZONE TEST

Glucose **Fructose** **Maltose** **Lactose**

No.	Test	Observation	Inference	Reaction
1	**Molisch's Test** 2-3 drops of beta-naphthol solution are added to 2ml of the test solution. Very gently add 1ml of Conc. H_2SO_4 along the side of the test tube..	A deep violet coloration is produced at the junction of two layers.	Presence of carbohydrates.	This is due to the formation of an unstable condensation product of beta-naphthol with furfural (produced by the dehydration of the carbohydrate).
2	**Iodine test** 4-5 drops of iodine solution are added to 1ml of the test solution and contents are mixed gently.	Blue colour is observed.	Presence of polysaccharide.	Iodine forms coloured adsorption complexes with polysaccharides.
3	**Fehling's test** About 2 ml of sugar solution is added to about 2 ml of Fehling's solution taken in a test-tube. It is then boiled for 10 min	A red precipitate is formed	Presence of reducing sugar	This is due to the formation of cuprous oxide by the reducing action of the sugar.

No	Test	Observation	Inference	Reaction
4	**Benedict's test** To 5 ml of Benedict's solution, add 1ml of the test solution and shake each tube. Place the tube in a boiling water bath and heat for 3 minutes. Remove the tubes from the heat and allow them to cool.	Formation of a green, red, or yellow precipitate	Presence of reducing sugars	If the saccharide is a reducing sugar it will reduce Copper [Cu] (11) ions to Cu(1) oxide, a red precipitate
5	**Barfoed's test** To 2 ml of the solution to be tested added 2 ml of freshly prepared Barfoed's reagent. Place test tubes into a boiling water bath and heat for 3 minutes. Allow to cool.	A deep blue colour is formed with a red ppt. settling down at the bottom or sides of the test tube.	Presence of reducing sugars. Appearance of a red ppt as a thin film at the bottom of the test tube within 3-5 min. is indicative of reducing mono-saccharide. If the ppt formation takes more time, then it is a reducing disaccharide.	If the saccharide is a reducing sugar it will reduce Cu (11) ions to Cu(1) oxide

No.	Test	Observation	Inference	Reaction
6	**Seliwanoff test** To 3ml of of Seliwanoff’s reagent, add 1ml of the test solution. Boil in water bath for 2 minutes.	A cherry red colored precipitate within 5 minutes is obtained. A faint red colour produced	Presence of ketoses [Sucrose gives a positive ketohexose test] Presence of aldoses	When reacted with Seliwanoff reagent, ketoses react within 2 minutes forming a cherry red condensation product Aldopentoses react slowly, forming the coloured condensation product.
7	**Bial's test** Add 3ml of Bial’s reagent to 0.2ml of the test solution. Heat the solution in a boiling water bath for 2 minutes.	A blue-green product A muddy brown to gray product	Presence of pentoses. Presence of hexoses.	The furfurals formed produces condensation products with specific colour.
8	**Osazone Test** Two two ml of the test solution, add 3ml of phenyl hydrazine hydrochloride solution and mix. Keep in a boiling water bath for 30mts. Cool the solution and observe the crystals under microscope.	Formation of beautiful yellow crystals of osazone Needle shaped crystals Hedgehog crystals Sunflower shaped crystals	Glucose/fructose Presence of lactose Presence of maltose	Reducing sugars forms ozazone on treating with phenylhydrazine

Benedict’s Test

- **Benedict’s Test** is used to test for simple carbohydrates. The **Benedict’s test** identifies reducing sugars (monosaccharide’s and some disaccharides), which have free functional groups.

- Benedict’s solution can be used to test for the presence of glucose in urine.

INDICATION

- Diabetes mellitus
- Renal glycosuria

Composition and Preparation of Benedict's Solution

- Benedict's solution is a deep-blue alkaline solution used to test for the presence of the aldehyde functional group, – CHO.

- *Anhydrous sodium carbonate(NA2CO3) = 100 gm*
  Sodium citrate – 173 gm
  Copper(II) sulfate pentahydrate = 17.3 gm

Procedure of Benedict’s Test

- Approximately 5 ml of sample is placed into a clean test tube.
- 2 ml (8 drops) of Benedict’s reagent ($CuSO_4$) is placed in the test tube.
- The solution is then heated in a boiling water bath for 3-5 minutes.
- Observe for color change in the solution of test tubes or precipitate formation.

- Reducing sugar in strong alkaline medium form enediols. Enediols are powerful reducing agents , reduces the metallic ions like cupric ions to cuprous ions

- Cu (OH) 2 $_{+}$ Sugar (enediols) $\xrightarrow{\text{boil}}$ CuO2 + Sugar acid

Results

positive benedict’s test: color change from blue to brick red ppt (glucose)
Negative Benedict’s test: no change in color(sucrose)

Color observed	Sugar %	Result interpretation
Blue	Nil	Absent of suagr
Green color	0.5%	+
Green ppt	0.5-1%	++
Yellow ppt	1-1.5%	+++
Orange ppt	1.5-2%	++++
Brick red ppt	>2%	+++++

Blue solution	Green / yellow ppt	Orange red ppt	Brick-red ppt
None	Traces of reducing sugar	Moderate	Large amount of reducing sugar

THANK YOU